Data Models in the Reddit Mobile Phone Application

CS-360 Assignment 2-2

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A key feature of mobile phone applications is its ability to pull data from various repositories, usually just its central servers, process the data, and present the results of the data processing to its users. Sometimes, the results are in the form of charts, and scientific figures and graphs. Phyphox (n.d.)[1] represented such a case, which was designed for scientists and engineers in mind, who represent a minority of the population. But mobile phone application development should also keep in mind the needs of a more general, non-technical audience[2] who use mobile phones for their everyday tasks, like communicating with their friends and family on social media platforms, "influencer-style" content creation, computer gaming,[3] sending emails, daily life activities that involve a calendar or "reminder tool," a step-count tracking tool, and other such casual applications.

For this paper, I intend to analyse the Reddit (n.d.) mobile phone application – which acts as a frontend to the Reddit platform.[4] This platform belongs to a kind of social networking platforms that allow end-users to share web links or their own generated content through a

[1] I have analysed Phyphox for a previous assignment: see Ahmann, A. (Sept. 6, 2026). *CS-360: Assignment 1-4: Assessment of "Phyphox" mobile phone application*.
[2] Scientists and engineers typically have more computer know-how when compared to the general population, even if they did not study computer science, information technology, or related fields.
[3] As I have learnt from previous experience with other mobile phone users, there even exists a mobile version of the *Call of Duty*® gaming franchise: https://play.google.com/store/apps/details?id=com.activision.callofduty.shooter
[4] This addresses the "choose a mobile app to analyze" requirement (CS-360 n.d.).

"self-post."[5] This Android app allows for many end-users to communicate with one-another, implying that by its very nature, it needs to request the relevant data that have been stored in large, non-volatile databases regarding a number of features linked to the end-user in question, like the "subreddit" communities that they follow and their user profile data.[6]

In order to assess the user interface for the Reddit mobile application, I installed it onto my Android phone and logged into a Reddit account that I have previously created. On the initial run of the app, a welcome screen is shown to the end-user, with the Reddit logo on the top of the phone's screen, a label (see zyBooks n.d., §2.6) reading "The most real place on the internet," a graphic depicting Reddit end-users, and finally two buttons (see zyBooks n.d., §2.5). After the end-user logs in, either by creating a new account or logging into a pre-existing account, they are presented with four layouts (see zyBooks n.d., §2.1) that present different kinds of data: "Home," "Create," "Inbox," and "You." A hamburger menu is also given in the top-left is also given.[7]

The "Home" layout group shows a feed of posts and other user-generated content that originated from their respective "subreddit" communities.[8] The "Create" layout group allows the end-user to submit their own links to external content, or their own self-generated content, into a subreddit that they are authorized to post to. The "Inbox" layout

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[5] This addresses a "briefly identify the app and explain its purpose" from CS-360 (n.d.).
[6] This part briefly touches on the requirement to "determine what data each component of the app either displays or accepts as input." I will elaborate on this further.
[7] I will not elaborate on its contents to keep this paper shorter.
[8] This paragraph explains the major user interface components, in accordance with the "List the major UI components on each screen" task from CS-360 (n.d.).

group pushes notifications onto the end-user, which typically involves other end-users responding to threads that the end-user is participating in, or private messages sent to the end-user from other end-users. Finally, the “You” layout group shows a rendering of the end-user’s “Reddit” user page.

Data comes in many forms: like integers, floating-point numbers, strings, characters, and even `structs` and `classes` that allow developers to create new abstract data types from primitives. In the case of Reddit, user-generated content comes in the form of posts and comments. The features[9] that embody these “posts” and “comments” come in the form of “upvotes,” “content body” (or link in the case of a post to an external online resource), and “date posted.”[10] In the case of a post, the external links and content that constitute a post, and the content that constitutes a comment, are both input from, and output to, the end-user. “Upvotes” are an aggregate of many end-users' approvals, where the input is taken from the end user pushing a “thumbs up” button, and the output is the sum of end-users who have pushed the “thumbs up” button. The ”date posted” feature is not directly input by the end-user but is worked out by the client and server components of the mobile app by services that keep track of the date and time. The data that Reddit works with is stored mainly on their central servers and data centres, and possibly on third-party servers and content delivery networks.

[9] Here, I address the “Identify where data is included in the app” directive and the “discuss the possible data sources of the mobile app” from CS-360 (n.d.).

[10] There are more features for the data that embody posts and comments, but I will just discuss these three bits of field to keep the paper simple.

The end-users interact primarily with data[11] in reference to user-generated content and shared links. Man is a social creature who prefers to work on activities, and work in groups, either directly or indirectly. In recent times, most of the human race has found itself in societies that are abundant in wealth,[12] which has enabled individuals to pursue more self-fulfilling kinds of work. Amongst such self-fulfillment comes in the form of content creation, where such content can be stored and processed by a global network of computer and information systems. Reddit solves the problem of allowing end-users to share, and generate, "creative works," and doing so as a community effort.

References

CS-360 (n.d.). *Module Two Assignment Guidelines and Rubric*. Southern New Hampshire University.

Phyphox (n.d.). *Physical Phone Experiments*. https://phyphox.org/

Reddit (n.d.). Apps on Google Play. Retrieved on Sept. 12, 2026, from: https://play.google.com/store/apps/details?id=com.reddit.frontpage

zyBooks (n.d.). *CS-360: Mobile Architecture and Programming [Sept. – Oct. 2026 Semester]*.

[11] This part addresses the "Explain how the data that users interact with helps them meet their goals" part of CS-360 (n.d.).

[12] The exception here are those hunter gatherer societies which focus on basic survival. Their problems are simple, but their wealth is limited when compared to fully developed nation-states.